# CSA5802 - DEVSECOPS ENGINEERING AND APPLICATION SECURITY

## CO2- ASSESSMENT TASK 2:

## EMAIL INFRASTRUCTURE SECURITY ASSESSMENT AND THREAT PROPAGATION ANALYSIS

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## 1. Introduction

Email is a critical communication system used by organizations for internal and external interactions. However, it is also one of the most targeted attack vectors for cyber threats such as phishing, malware, and data breaches. This report presents a comprehensive assessment of email infrastructure security and analyzes how threats propagate within such systems.

## 2. Objectives

- To analyze the architecture of email infrastructure
- To identify vulnerabilities in email systems
- To assess security mechanisms and controls
- To study threat propagation techniques in email networks
- To recommend mitigation strategies

## 3. Email Infrastructure Overview

An email system consists of several key components:

### 3.1 Mail Servers

- **SMTP (Simple Mail Transfer Protocol):** Sends emails
- **POP3/IMAP:** Retrieves emails

### 3.2 Mail Transfer Agents (MTAs)

- Responsible for routing emails between servers

### 3.3 Mail User Agents (MUAs)

- Email clients like Gmail, Outlook

### 3.4 DNS and MX Records

- Direct emails to appropriate mail servers

## 4. Security Mechanisms in Email Systems

### 4.1 Authentication Protocols

- SPF (Sender Policy Framework)
- DKIM (DomainKeys Identified Mail)
- DMARC (Domain-based Message Authentication, Reporting & Conformance)

### 4.2 Encryption

- TLS (Transport Layer Security)
- End-to-End Encryption (PGP, S/MIME)

### 4.3 Spam Filtering

- Detects unwanted or malicious emails

### 4.4 Anti-Malware Systems

- Scans attachments and links

## 5. Vulnerability Assessment

### 5.1 Common Vulnerabilities

- Weak authentication mechanisms
- Misconfigured mail servers
- Lack of encryption
- Outdated software

### 5.2 Human Factors

- Phishing susceptibility
- Poor password practices

### 5.3 System-Level Issues

- Open relay servers
- Improper access controls

## 6. Threat Propagation Analysis

Threat propagation refers to how an attack spreads through email systems.

### 6.1 Phishing Attacks

- Attackers impersonate trusted entities
- Users unknowingly share credentials

### 6.2 Malware Distribution

- Malicious attachments or links

- Once opened, malware spreads across systems

### 6.3 Email Spoofing

- Fake sender addresses used to deceive users

### 6.4 Lateral Movement

- Compromised accounts send emails internally
- Attack spreads within the organization

### 6.5 Business Email Compromise (BEC)

- Attackers target executives or finance departments
- Leads to financial fraud

## 7. Risk Assessment

| Threat Type | Impact | Likelihood | Risk Level |
|---|---|---|---|
| Phishing | High | High | Critical |
| Malware | High | Medium | High |
| Spoofing | Medium | High | High |
| BEC | Very High | Medium | Critical |

## 8. Mitigation Strategies

### 8.1 Technical Controls

- Implement SPF, DKIM, DMARC
- Use strong encryption (TLS)
- Deploy advanced email filtering systems

### 8.2 User Awareness

- Conduct phishing awareness training
- Promote strong password practices

### 8.3 Monitoring and Detection

- Use SIEM tools
- Monitor suspicious login activity

### 8.4 Incident Response

- Immediate isolation of compromised accounts
- Regular backups and recovery plans

## 9. Best Practices

- Enable multi-factor authentication (MFA)
- Regularly update and patch systems
- Restrict access based on roles
- Perform regular security audits

## 10. Conclusion

Email infrastructure is a major target for cyber threats due to its widespread use. A strong combination of technical controls, user awareness, and continuous monitoring is essential to prevent attacks and limit threat propagation. Organizations must adopt proactive security measures to ensure the integrity and confidentiality of email communications.